ECE2191 Cheat Sheet

Probability and AI for Engineers. Methods, conditions, traps and past-paper answers. Use with the formula sheet and Z-table. From your weekly notes, practice papers 1–6, the mid-sem, the 2026 formula sheet and the Z-table. Worked numbers rechecked in Python.

1 How to score, and what is handed to you

Read once before you start.

Every answer: formula → substitution → number → one sentence of meaning

Trap: "No marks if no reasoning." Text boxes: a/b, a^(b), C(n,r), sqrt(), Phi(x).

Formula sheet gives: norms, inner product, $W = (X^TX)^{-1}X^TY$, MSE, R^2, BIC, GD update, sigmoid, CE, softmax, activations, losses, quadratic formula, derivative rules

Trap: It gives NO probability formulas. Sections 2 to 8 below are what you must bring.

Not on the formula sheet: $\sigma' = \sigma(1-\sigma)$, ridge, precision/recall/F1, X^+ for fat X, C(n+k,k)

Z-table gives $\Phi(z) = P(Z \le z)$, 4 decimals, row = 1st decimal, column = 2nd

Needs: $Q(z) = P(Z>z) = 1-\Phi(z)$. $\Phi(-z) = 1-\Phi(z)$.

Trap: Older papers gave a Q-table. The 2026 table is Φ.

$\Phi(1)=0.8413$ $\Phi(1.5)=0.9332$ $\Phi(1.6)=0.9452$ $\Phi(1.96)=0.9750$ $\Phi(2)=0.9772$ $\Phi(2.576)\approx0.995$ $\Phi(0.84)=0.7995$

z for a two-sided CI: 90% 1.645, 95% 1.96, 99% 2.576

2 Which tool? Read the question, pick the section

Seats, committees, "exactly k faulty in a batch" → 3. Event trees, "given that" → 3

Fixed n trials → Binomial. Until first success → Geometric. Count per time → Poisson. Wait time → Exponential (4)

PDF with unknown constant, CDF, E, Var, Y=aX+b → 5

Joint table or joint PDF, independent?, uncorrelated?, conditional → 6

Gaussian "probability exceeds", sample mean, CLT → 7. CI, sample size → 8. Decide 0/1 from noisy y → 9

Fit, residuals, R^2, BIC, features → 10. Learning rate, iterations → 11. Classes, metrics, loss → 12. ReLU nets, ridge → 13

Counting rule, E/Var rules, sums of RVs, tails, integrals, models, derivatives, diagnosis → 15

3 Probability rules and counting

practice 2 Q6, 3 Q3, 4 Q1, Q6

$P(A|B) = P(A\cap B)/P(B)$. $P(A\cap B) = P(A|B)P(B)$

Trap: $P(A|B) \neq P(B|A)$.

Total: $P(A) = \Sigma\, P(A|B_i)P(B_i)$. Bayes: $P(B_i|A) = P(A|B_i)P(B_i)/P(A)$

Needs: B_i partition the space.

Trap: Draw the tree. Leaves sum to 1.

Independent $\Leftrightarrow$ $P(A\cap B) = P(A)P(B)$

$C(n,k) = n!/(k!(n-k)!)$. Arrangements in a line n!. Round table $(n-1)!$

Hypergeometric: P(j special in k drawn from N with M special) = C(M,j)C(N−M,k−j)/C(N,k)

Needs: No replacement.

Groups by category: Π C(category size, chosen) / C(total, group size)

Trap: Chosen counts must add to the group size.

Past paper: Spouse next to me at a 20-seat round table: 2·18!/19! = 2/19. Exactly 1 of 5 chips faulty (11 of 30 broken): C(11,1)C(19,4)/C(30,5) = 0.2992.

Past paper: P(not faulty) = 0.8(5/30)+0.6(25/30) = 0.6333; P(new | not faulty) = 0.1333/0.6333 = 0.2105.

4 Discrete distributions

practice 2 Q1, 3 Q2, 4 Q4, mid-sem Q1–2

Model	Use when	P(X=k) or f(x)	E	Var
Bernoulli(p)	one yes/no trial	p, 1−p	p	p(1−p)
Binomial(n,p)	fixed n, independent, same p	$C(n,k)p^k(1-p)^{n-k}$	np	np(1−p)
Geometric(p)	trials until 1st success	$(1-p)^{k-1}p$, k≥1	1/p	$(1-p)/p^2$
Poisson(λ)	count per interval, rate given	$e^{-\lambda}\lambda^k/k!$	λ	λ
Hypergeom.	draw k without replacement	C(M,j)C(N−M, k−j)/C(N,k)	kM/N	—
Uniform(a,b)	equally likely in a range	1/(b−a)	(a+b)/2	$(b-a)^2/12$
Exponential(λ)	waiting time, constant rate	$\lambda e^{-\lambda x}$	1/λ	$1/\lambda^2$
Gaussian(μ,σ²)	noise, many small effects	$e^{-(x-\mu)^2/2\sigma^2}/(\sigma\sqrt{2\pi})$	μ	σ^2

Memoryless: only Geometric (discrete) and Exponential (continuous)

Binomial(n,p): $P(k) = C(n,k)p^k(1-p)^{n-k}$, E = np, Var = np(1−p)

Needs: Fixed n, independent, same p.

Trap: P(at least one) = $1-(1-p)^n$. Bursty errors ⇒ not binomial, but E[S] = np still holds.

Geometric(p): $P(k) = (1-p)^{k-1}p$, $k \ge 1$. $P(X>k) = (1-p)^k$. E = 1/p, Var = $(1-p)/p^2$

Needs: Trials up to and including first success.

Trap: Memoryless: 5 failures done, P(more than 3 more) = 0.8^3.

Poisson(λ): $P(k) = e^{-\lambda}\lambda^k/k!$, E = Var = λ

Needs: λ = rate × interval (same units).

Trap: Scale λ first: 600/min over 200 ms ⇒ λ = 2. 1460/yr over a day ⇒ λ = 4.

$P(N=0) = e^{-\lambda}$. $P(N \le 2) = e^{-\lambda}(1+\lambda+\lambda^2/2)$

Trap: λ=2: 0.1353, 0.6767. λ=4: 0.0183, 0.2381.

$E[X] = \Sigma x\, p(x)$. $Var = E[X^2] - (E[X])^2$

Needs: PMF sums to 1.

Y = g(X) discrete: P(Y=y) = sum of P over the x values (or regions) that map to y

Past paper: Frisbee (p=0.1): $P(X=k)=0.9^{k-1}(0.1)$; $P(X>3)=0.9^3=0.729$; E=10. Meatballs in 100 throws: Binomial(100, 0.1).

5 Continuous: PDF, CDF, unknown constant

practice 2 Q3, 3 Q4, 4 Q2, mid-sem Q4–5, Q13–14

$\int f = 1$ fixes the constant. A second fact (e.g. E[X]) fixes a second constant

F(x) = ∫ from the lower end to x of f. Write 0 below and 1 above

Needs: Piecewise answer.

Trap: P(X = x) = 0 for continuous X.

P(a<X<b) = F(b)−F(a). $E[X] = \int xf$. $E[X^2] = \int x^2 f$

Trap: Symmetric f ⇒ E = centre, P(X ≥ centre) = ½.

Y = aX+b: E = aE[X]+b, $Var = a^2Var(X)$

Trap: °F = 1.8°C+32 ⇒ $Var_F = 3.24\, Var_C$. Y = 2X+1 ⇒ Var ×4.

Exponential(λ): $f = \lambda e^{-\lambda x}$, $F = 1-e^{-\lambda x}$, $P(X>x) = e^{-\lambda x}$, E = 1/λ, $Var = 1/\lambda^2$

Needs: Constant failure rate.

Trap: Mean 200 h ⇒ λ = 0.005, Var = 40,000, SD = 200. "Most fail near 200 h" is FALSE: density is largest at 0.

Gaussian N(μ,σ²): $f = e^{-(x-\mu)^2/2\sigma^2}/(\sigma\sqrt{2\pi})$, peak $1/(\sigma\sqrt{2\pi})$

Trap: Higher peak ⇒ smaller σ. f(0) is not a probability.

Flat CDF on an interval ⇒ zero probability there

Trap: Not uniform.

Past paper: f = a(4−(x−38)²) on [36,40] ⇒ a = 3/32, E = 38, P(X ≥ 38) = ½. f = ax^b on [0,1], E=0.6 ⇒ a=1.5, b=0.5, F = $x^{1.5}$, Var = 3/7−0.36 = 0.0686, P(X>0.5) = 0.646.

6 Joint distributions, covariance, vectors

practice 3 Q5, 4 Q5–6, mid-sem Q3, Q15

Marginal: $p_X(x) = \Sigma_y\, p(x,y)$, $f_X(x) = \int f(x,y)dy$

Trap: Limits for y may depend on x.

Conditional: $f(y|x) = f(x,y)/f_X(x)$

Trap: f=2 on 0<y<x<1 ⇒ f_X=2x, f(y|x)=1/x, P(Y<x/2 | X=x) = ½.

Claim	Test	Implies
Independent	$p(x,y) = p_X\, p_Y$ for EVERY cell (support a rectangle)	uncorrelated
Uncorrelated	Cov = E[XY] − E[X]E[Y] = 0	NOT independence (unless jointly Gaussian)
Orthogonal	E[XY] = 0	uncorrelated only if a mean is 0

$Cov = E[XY] - E[X]E[Y]$. $\rho = Cov/(\sigma_X\sigma_Y) \in [-1,1]$

Trap: Cov = 0 ⇒ uncorrelated, NOT independent (X on {−1,0,1}, Y = X^2).

E[aX+bY] = aE[X]+bE[Y] (always). Var(X+Y) = VarX + VarY + 2Cov

Trap: Cov(F, aF+b) = a Var(F).

Vector: $C = E[(X-\mu)(X-\mu)^T] = R - \mu\mu^T$, symmetric PSD

Needs: Diagonal = variances.

Jointly Gaussian and Cov = 0 ⇒ independent

Needs: Both needed.

Past paper: p(0,1)=p(1,0)=p(2,1)=1/3: E[X]=1, E[Y]=2/3, E[XY]=2/3 ⇒ Cov = 0, uncorrelated. P(0,0)=0 ≠ (1/3)(1/3) ⇒ dependent.

Past paper: 3 balls from 2R,3W,4B: p(x,y) = C(2,x)C(3,y)C(4,3−x−y)/C(9,3).

7 Gaussians, sample means, LLN, CLT

practice 2 Q2, 3 Q1, Q6, 4 Q7, mid-sem Q6–8

$Z = (X-\mu)/\sigma$. $P(X \le x) = \Phi(z)$. $P(X > x) = 1-\Phi(z)$

Trap: Divide by σ, not σ^2.

$\bar{X} \sim N(\mu, \sigma^2/n)$: use $SE = \sigma/\sqrt{n}$. Sum $S_n \sim N(n\mu, n\sigma^2)$

Needs: Gaussian data, or CLT for large n.

Trap: A total uses the sum form.

CLT: IID, finite mean and variance; n ≥ 30 is a rule of thumb

Trap: Rare events (p = 0.001, n=30): $P(\bar{X}=0) = 0.97$ ⇒ not Gaussian.

LLN: $\bar{X} \to \mu$ (where). CLT: shape around μ
Needs: IID needed.
Trap: Drift or bursts break IID: more data does not fix it.

Symmetric interval with prob p: $\mu \pm z{\cdot}\sigma/\sqrt{n}$ with $\Phi(z) = (1+p)/2$
Trap: 95% ⇒ 1.96. 60% ⇒ $\Phi(z)=0.8$ ⇒ z = 0.84.

Past paper: Sources above 240: A (230,5) z=2 ⇒ 0.0228; B (225,10) z=1.5 ⇒ 0.0668 ⇒ B more likely. Sensor lifetimes ≥30,000: S1 0.0228, S2 0.0668 ⇒ choose S2.

Past paper: Mid-sem Q8: (13.2−12)/(6/√64)=1.6 ⇒ 1−0.9452=0.0548. Chef: σ=10, n=16 ⇒ SE 2.5; P(45<$\bar{X}$<52)=Φ(2.8)−0.5=0.4974; 95% of means in 45±4.9.

8 Estimation, confidence intervals, sample size

practice 2 Q4–5, mid-sem Q9–12

$\bar{x}$ estimates μ. $s^2 = \Sigma(x_i-\bar{x})^2/(n-1)$
Trap: Divide by n−1.

Unbiased: $E[\hat{\theta}] = \theta$. Consistent: $\hat{\theta} \to \theta$ as n grows
Trap: X_1 and $\bar{X}$ are both unbiased; $\bar{X}$ has the smaller SE.

Situation	Use	Conditions
P(one reading > t)	$1-\Phi((t-\mu)/\sigma)$	X Gaussian
P(sample mean > t)	$1-\Phi((t-\mu)/(\sigma/\sqrt{n}))$	Gaussian data or large n (CLT)
P(total > t)	$1-\Phi((t-n\mu)/(\sigma\sqrt{n}))$	same
CI for μ	$\bar{x} \pm z{\cdot}\sigma/\sqrt{n}$	IID, σ known
σ unknown	$\bar{x} \pm t{\cdot}s/\sqrt{n}$ (Student t)	z is not exact
Sample size	$n = (z\sigma/E)^2$, round up	σ known

Meaning: 95% of intervals built this way contain μ
Trap: Not "95% of items" and not "μ has 95% probability".

CI fully inside a tolerance ⇒ supports "mean is within tolerance"

Higher confidence ⇒ wider. Larger n ⇒ narrower

$n = (z\,\sigma\,/\,E)^2$, ALWAYS round up
Needs: E = maximum margin of error.
Trap: Halve E ⇒ 4× n.

Past paper: P(|$\bar{X}$−μ| < σ/10) ≥ 0.95 ⇒ 1.96σ/√n ≤ σ/10 ⇒ n ≥ 384.16 ⇒ n = 385.

9 Binary detection in Gaussian noise

practice 4 Q3, practice 5

Decide "0" (X=−1) if $f(y|-1)P_0 > f(y|+1)P_1$
Needs: Y = X + N, N ~ $N(0,\sigma^2)$.

Threshold $T = (\sigma^2/2)\ln(P_0/P_1)$: decide 0 if y < T
Needs: Equal noise, levels ±1.
Trap: Equal priors ⇒ T = 0.

$P(\text{error}) = P_0{\cdot}Q((T+1)/\sigma) + P_1{\cdot}Q((1-T)/\sigma)$
Trap: Q = 1−Φ.

Past paper: $P_0 = 2P_1$, $\sigma^2 = 1/16$: T = ln2/32 = 0.0217; P(error) ≈ 3.0×10^{-5}.

10 Regression

practice 5, 6 Q1–11

Univariate: $w = \hat{\sigma_{xy}}/\hat{\sigma_{xx}}$, $b = \bar{y} - w\,\bar{x}$ (or $w = (\Sigma xy - m\,\bar{x}\bar{y})/(\Sigma x^2 - m\,\bar{x}^2)$)
Needs: Divide by m in $\hat{\sigma}$.

Least squares with bias ⇒ $\Sigma e_i = 0$, $\Sigma x_i e_i = 0$, $\Sigma \hat{y}_i e_i = 0$
Needs: $e_i = y_i - \hat{y}_i$.
Trap: Use these to recover lost residuals.

MSE = $(1/m)\Sigma e_i^2$. $R^2 = 1 - SSE/SST$ (fraction of variance explained)
Trap: Higher R^2 is better on training data, but always grows with more features.

Tall X (m ≥ n): $W = (X^TX)^{-1}X^TY$. Fat X: $w = X^T(XX^T)^{-1}y$ (minimum norm)
Needs: Add a column of 1s for the bias.
Trap: Fat: infinitely many zero-error fits; zero training error means little.

Linear in parameters ⇒ still linear regression (x^2, cos x, sin x features)

Polynomial features of degree ≤ k from n inputs: C(n+k, k)
Trap: n=3: k=2 ⇒ 10, k=3 ⇒ 20. 20 params with 12 samples ⇒ underdetermined.

BIC = d ln m + m ln MSE: lower is better
Trap: Can be negative. Only the minimum matters.

Feature rescaling: LS predictions unchanged; ridge changes
Trap: x′=2x−1, y′=10y+5 ⇒ w′=5w, R^2 same, MSE ×100.

Fit y on x and x on y: $w_1 w_2 = R^2$
Trap: Lines coincide only if $R^2 = 1$.

Outliers pull the line; check residuals, remove bad data or refit

Past paper: BIC, m=50: A (d=3, MSE 0.40) = −34.08, B (d=6, 0.33) = −31.96 ⇒ A. B wins only if MSE < 0.316. With m=5000 ⇒ B.

Past paper: Data x=1..5, y=2,3,5,4,6: $\hat{y} = 1.3 + 0.9x$; $w_1 w_2 = 0.81 = R^2$.

11 Gradient descent

practice 5, 6 Q12–16

$w \leftarrow w - \eta\nabla L$; stop when $\|\nabla L\|$ small, loss change small, or max iterations
Needs: η > 0.

Quadratic $f = (\lambda/2)(w-w^*)^2 \Rightarrow w^{(t)}-w^* = (1-\eta\lambda)^t(w^{(0)}-w^*)$
Trap: Converges iff $|1-\eta\lambda| < 1$, i.e. $0 < \eta < 2/\lambda$. Best one-step $\eta = 1/\lambda$.

Several curvatures: $\eta < 2/\lambda_{max}$
Trap: $\frac{1}{2}(w_1^2+9w_2^2) \Rightarrow 0 < \eta < 2/9$.

η too small: slow. Too big: oscillates or diverges. Plateau: decay η
Trap: Faster smooth drop after a change ⇒ η was increased.

$\sigma' = \sigma(1-\sigma)$. Chain rule: $dy/dw_1 = (dy/d\theta_2)(d\theta_2/d\theta_1)(d\theta_1/dw_1)$

Past paper: $f=(w-2)^2+1$, $w_0=5$, $w_1=3.8$ ⇒ η=0.2; $w_2=3.08$, $w_3=2.648$; |w−2| <0.01 needs t=12. η=1 oscillates 5, −1.

Past paper: $f=\frac{1}{2}x^2-4x+12$ from $x_0=-2$ ⇒ η=1 lands on x=4. $\nabla[(x_1-3)^3\sigma(x_2)]$ at (−1,1) = (35.09, −12.58).

12 Classification

practice 5, 6 Q17–22

Logistic: $\hat{y} = \sigma(w^Tx)$, class 1 if $\hat{y} \geq 0.5 \Leftrightarrow w^Tx \geq 0$
Needs: Boundary is a line or hyperplane.

BCE: $-y\ln\hat{y} - (1-y)\ln(1-\hat{y})$. Gradient $(\hat{y}-y)x$
Needs: Convex.
Trap: Largest $|\hat{y}-y|$ ⇒ largest update. MSE + sigmoid stalls.

Softmax $p_r = e^{z_r}/\Sigma e^{z_i}$; CCE = −ln p_true
Needs: One-hot labels.
Trap: $z_i/\Sigma z$ fails with negatives. Integer labels imply false order.

Only logistic allowed, K classes ⇒ K one-vs-rest models
Trap: 3 classes ⇒ 3 models.

Boundary $w_1x_1+w_2x_2+b = 0$; w points to class +1
Trap: Plug a point (origin) to find the side.

Metric	Formula	Choose it when
Accuracy	(TP+TN)/N	classes balanced
Precision	TP/(TP+FP)	false alarms are costly
Recall	TP/(TP+FN)	misses are dangerous (lower the threshold)
F_1	2PR/(P+R)	imbalanced classes

Hinge max(0,1−yf): y=−1 minimised for f ≤ −1. Log loss $\ln(1+e^{-yf})$: no finite minimiser

Past paper: Softmax (5.2, 3.6, −2.7) ⇒ (0.832, 0.168, 0.0003). TP=6, FP=2, FN=3 ⇒ P=0.75, R=0.667, F_1=0.706.

13 Neural nets and ridge

practice 6 Q23+

Neuron $\varphi(w^Tx)$; with σ it is logistic regression

$h = \varphi(W_1{}^Tx)$, $y = W_2{}^Th$. Params ≈ nH + Hp (+ biases)
Trap: 100·500 + 500·1 = 50,500.

ReLU [0,∞), sigmoid (0,1), tanh (−1,1)

No activations ⇒ the whole MLP collapses to one linear map

ReLU(x) − 2ReLU(x−1) + ReLU(x−2): triangle, peak 1 at x=1, zero outside [0,2]
Needs: One kink per hidden unit.

Ridge: $L = MSE + \alpha\|w\|^2$, $w = (X^TX + \alpha I)^{-1}X^Ty$
Needs: α ≥ 0.
Trap: Larger α shrinks w. Sensitive to feature units.

14 Concept traps (mid-sem multiple choice)

mid-sem

Linearity of expectation needs no independence; variance additivity does

Uncorrelated but dependent is possible ($Y = X^2$)

Larger PDF peak ⇒ smaller σ; P(A=0) = 0 for continuous A

n = 30 is not automatically enough for the CLT

Calibration drift breaks the LLN conclusion

Using s instead of σ ⇒ t-distribution

Halving the margin of error ⇒ quadruple n

15 Reference tables for any question

Standard results that work on any question, not only past papers.

Choose k from n	With replacement	Without replacement
Order matters	n^k	n!/(n−k)!
Order ignored	C(n+k−1, k)	C(n, k)

Line of n: n!. Round table: (n−1)!. Repeated items: $n!/(n_1!n_2!\ldots)$

Rule	Formula	Needs
Shift and scale	$E[aX+b] = aE[X]+b$; $Var(aX+b) = a^2Var(X)$	always
Sum	$E[X+Y] = E[X]+E[Y]$	always
Variance of sum	$Var(X+Y) = VarX + VarY + 2Cov(X,Y)$	always; $Cov = 0$ if independent
Covariance	$Cov(aX+b, cY+d) = ac \cdot Cov(X,Y)$; $Cov(X,X) = VarX$	always
Product	$E[XY] = E[X]E[Y]$	independent (or uncorrelated)
Function	$E[g(X)] = \Sigma\, g(x)p(x)$ or $\int g(x)f(x)dx$	always
Sample mean	$E[\bar{X}] = \mu$; $Var(\bar{X}) = \sigma^2/n$	IID for the variance

Sum of independent	Result
n Bernoulli(p)	Binomial(n, p)
Binomial(n,p) + Binomial(m,p)	Binomial(n+m, p)
Poisson(λ_1) + Poisson(λ_2)	Poisson($\lambda_1+\lambda_2$)
N(μ_1,σ_1^2) + N(μ_2,σ_2^2)	N($\mu_1+\mu_2$, $\sigma_1{}^2+\sigma_2{}^2$)
n IID, any distribution	$\approx N(n\mu, n\sigma^2)$ for large n (CLT)

Model	Tail or CDF
Geometric	$P(X > k) = (1-p)^k$
Binomial	$P(X \geq 1) = 1 - (1-p)^n$; otherwise sum terms
Poisson	$P(X \leq k) = e^{-\lambda} \Sigma_{i=0}{}^{k}\, \lambda^i/i!$
Uniform(a,b)	$F(x) = (x-a)/(b-a)$
Exponential	$F(x) = 1 - e^{-\lambda x}$; $P(X > x) = e^{-\lambda x}$
Gaussian	$F(x) = \Phi((x-\mu)/\sigma)$

$\int_0{}^{\infty} x^n e^{-\lambda x} dx = n!/\lambda^{n+1}$. $\int_a{}^{b} x^n dx = (b^{n+1}-a^{n+1})/(n+1)$. $\int e^{-x^2/2} dx = \sqrt{(2\pi)}$
Needs: Use for E and Var of continuous models.

$\Sigma_{k\geq 0}\, r^k = 1/(1-r)$. $\Sigma_{k\geq 1}\, k\, r^{k-1} = 1/(1-r)^2$. $\Sigma_{k\geq 0}\, \lambda^k/k! = e\text{^}\lambda$
Needs: $|r| < 1$.

Model	Output	Loss	Solved by	Boundary / shape
Linear regression	w^Tx	MSE	$(X^TX)^{-1}X^Ty$	hyperplane
Polynomial / basis	$w^T\varphi(x)$	MSE	same, on $\varphi(x)$	curve
Ridge	w^Tx	MSE + $\alpha\|w\|^2$	$(X^TX+\alpha I)^{-1}X^Ty$	hyperplane, smaller w
Logistic	$\sigma(w^Tx)$	BCE	gradient descent	linear boundary
Softmax	softmax(W^Tx)	CCE	gradient descent	piecewise linear
Shallow NN	$W_2{}^T\varphi(W_1{}^Tx)$	MSE or CE	GD + backprop	nonlinear

Function	Derivative / gradient
$\sigma(z)$	$\sigma(1-\sigma)$
tanh(z)	$1 - \tanh^2(z)$
ReLU(z)	1 if $z > 0$, else 0
w^Tx	x
$\|w\|^2$	2w
$(1/m)\|Xw - y\|^2$	$(2/m)X^T(Xw - y)$
BCE with $\sigma(w^Tx)$	$(\hat{y} - y)x$
CCE with softmax, w.r.t. z	$\hat{y} - y$

Train error	Test error	Diagnosis	Fix
high	high	underfit	more features or degree, less regularisation
low	high	overfit	more data, lower degree, ridge, pick by BIC
low	low	good fit	—

Loss curve	Cause	Fix
rises or explodes	η too large	reduce η
zig-zags, plateaus above minimum	η a bit large	decay η
falls very slowly	η too small	increase η
smooth and fast to the minimum	η well chosen	—

ECE2191 Formulae Sheet 2026

AI formula and theorems

1 Vector Operations.

Vectors are foundational elements of a structured algebra known as a vector space, which allows for the linear combination of any two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ of identical dimensions

Table 1: Common vector operations

Function	Definition	Notes
Vector norms	$\|\boldsymbol{x}\|_p \triangleq \sqrt[p]{\sum_{i=1}^{n} \left\|x^p[i]\right\|}$	Finds the norm of a vector
Inner product	$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \boldsymbol{u}^\top \boldsymbol{v} = \sum_{i=1}^{n} u[i]v[i]$	Also known as dot product
Cosine similarity	$\cos(\theta) = \frac{\langle \boldsymbol{u}, \boldsymbol{v} \rangle}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$	θ is the angle between vectors $\boldsymbol{u}$ and $\boldsymbol{v}$

2 Multivariate Regression.

In the context of multivariate regression, the variables $\boldsymbol{X}$ and $\boldsymbol{Y}$ can be described as follows:

- $\boldsymbol{X}$: The feature matrix of dimension $m \times n$, where n is the number of features and m is the number of samples.
- $\boldsymbol{Y}$: The target matrix of dimension $m \times p$, where p is the number of target variables.

Note: The dimensions mentioned assume that $\boldsymbol{X}$ and $\boldsymbol{Y}$ are matrices where rows correspond to samples.

$$\boldsymbol{W} = \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{Y} \, .$$

3 Regression Tools.

The following are common tools and evaluation metrics used in regression models. Let m denote the number of data points, $\boldsymbol{x}_i \in \mathbb{R}^d$ the feature vector (with d input dimensions), and $\boldsymbol{W}^*$ the estimated model parameters.

$$\texttt{MSE}(\boldsymbol{W}^*) = \frac{1}{m} \sum_{i=1}^{m} \|\boldsymbol{y}_i - \boldsymbol{W}^{*\top} \boldsymbol{x}_i\|^2$$

The **mean squared error (MSE)** is a common loss function used in regression tasks.

$$R^2 \overset{\text{eq}}{=} 1 - \frac{\sum\limits_{i=1}^{m} \|\boldsymbol{y}_i - \boldsymbol{W}^{*\top} \boldsymbol{x}_i\|^2}{\sum\limits_{i=1}^{m} \|\boldsymbol{y}_i - \bar{\boldsymbol{y}}\|^2}$$

MONASH University

The **coefficient of determination** R^2 shows the goodness of fit for a regression model based on the estimated outputs

$$\texttt{BIC}(\mathcal{D}, \boldsymbol{W}^*) = d\log(m) + m\log\left(\texttt{MSE}(\boldsymbol{W}^*)\right)$$

The **Bayesian Information Criterion (BIC)** penalizes model complexity by incorporating the number of parameters d. Lower BIC values indicate a more parsimonious (better-balanced) model.

4 Gradient Descent.

Gradient descent allows us to iteratively update the weights of a model by finding the derivative of the loss function

$$\boldsymbol{\omega}^{(k)} = \boldsymbol{\omega}^{(k-1)} - \eta\nabla_{\boldsymbol{\omega}^{(k-1)}} f$$

5 Logistic regression.

The sigmoid function is typically used for binary classification problems

$$\sigma(x) = \frac{1}{1+\exp(-x)}$$

The cross entropy loss function is defined as (note we use $\log$ to denote the natural logarithm):

$$\mathcal{L}_{\text{CE}}(\boldsymbol{w}, \mathcal{D}) = -\frac{1}{m}\sum_{i=1}^{m}\left\{y_i\log(\sigma(\boldsymbol{w}^\top\boldsymbol{x}_i)) + (1-y_i)\log(1-\sigma(\boldsymbol{w}^\top\boldsymbol{x}_i))\right\}$$

The softmax function that is used for multi-class classification is defined as:

$$\boldsymbol{p} = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_K \end{pmatrix} = \frac{1}{\sum_{i=1}^{K}\exp(z_i)}\begin{pmatrix} \exp(z_1) \\ \exp(z_2) \\ \vdots \\ \exp(z_K) \end{pmatrix}$$

MONASH University

6 Function

Table 2: Common functions used in neural networks

Function	Definition	Notes
Sign	$\text{sign}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$	Output is -1, or 1
Sigmoid	$\sigma(x) = \frac{1}{1+\exp(-x)}$	Output in range $[0, 1]$
Tanh	$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{\exp(x)-\exp(-x)}{\exp(x)+\exp(-x)}$	Output in range $[-1, 1]$
ReLU	$\text{ReLU}(x) = \max(0, x)$	Non-negative output
Softmax	$z_i = \frac{\exp(x_i)}{\sum_{j=1}^{K} \exp(x_j)}$	Input x is K-dimensional, output z is K-dimensional; i-th element as in formula

Table 3: Common loss functions used in neural networks

Function	Formula	Notes
Mean Squared Error (MSE)	$\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$	y_i is true value, $\hat{y}_i$ is predicted
Binary Cross-Entropy (BCE)	$-y\log(\hat{y}) - (1-y)\log(1-\hat{y})$	y is either 0 or 1, $\hat{y}$ is predicted probability
Negative Log-Likelihood (NLL) (aka Categorical Cross-Entropy (CCE))	$-\sum_{i=1}^{K} y_i \log(\hat{y}_i)$	y_i is true class label, $\hat{y}_i$ is predicted probability y_i can be elements of a onehot vector.

Table 4: Trigonometric functions

Function	Definition	Notes
$\tan(x)$	$\frac{\sin(x)}{\cos(x)}$	Tangent
$\cot(x)$	$\frac{\cos(x)}{\sin(x)}$	Cotangent
$\sec(x)$	$\frac{1}{\cos(x)}$	Secant
$\csc(x)$	$\frac{1}{\sin(x)}$	Cosecant

Table 5: Hyperbolic functions

Function	Definition	Notes
$\sinh(x)$	$\frac{\exp(x)-\exp(-x)}{2}$	Hyperbolic sine
$\cosh(x)$	$\frac{\exp(x)+\exp(-x)}{2}$	Hyperbolic cosine
$\tanh(x)$	$\frac{\exp(x)-\exp(-x)}{\exp(x)+\exp(-x)}$	Hyperbolic tangent

7 Quadratic Function

A quadratic equation of the form

MONASH University

$$ax^2 + bx + c = 0, \qquad a \neq 0 \tag{1}$$

has the following solutions:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{2}$$

where the discriminant is defined as

$$\Delta = b^2 - 4ac \tag{3}$$

Interpretation:

- $\Delta > 0$: two distinct real roots,
- $\Delta = 0$: one repeated real roots,
- $\Delta < 0$: no real roots (two complex conjugate roots).

8 Derivatives

Table 6: Generic differentiation rules

Rule	Derivative
Sum/Difference	$\big(f(x) \pm g(x)\big)' = f'(x) \pm g'(x)$
Constant Multiple	$\big(cf(x)\big)' = cf'(x)$
Product	$\big(f(x)g(x)\big)' = f'(x)g(x) + f(x)g'(x)$
Quotient	$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - g'(x)f(x)}{[g(x)]^2}$
Chain rule	$f\big(g(x)\big)' = f'\big(g(x)\big)g'(x)$

Table 7: Common derivatives of elementary functions

Function	Derivative
k (constant)	0
x^n	nx^{n-1}
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\exp(x)$	$\exp(x)$
$\log(x)$	$\frac{1}{x}$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$
$\cot(x)$	$-\csc^2(x)$
$\sec(x)$	$\sec(x)\tan(x)$
$\csc(x)$	$-\csc(x)\cot(x)$

MONASH University

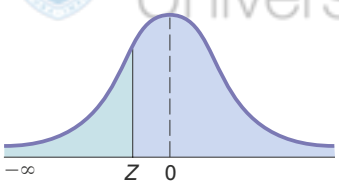
Table 8: Common hyperbolic functions and their derivatives

Function	Derivative
$\sinh(x)$	$\cosh(x)$
$\cosh(x)$	$\sinh(x)$
$\tanh(x)$	$\text{sech}^2(x)$

MONASH University

TABLE E.2
The Cumulative Standardized Normal Distribution

Entry represents area under the cumulative standardized normal distribution from $-\infty$ to Z



Cumulative Probabilities										
Z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
−6.0	0.000000001									
−5.5	0.000000019									
−5.0	0.000000287									
−4.5	0.000003398									
−4.0	0.000031671									
−3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
−3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
−3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
−3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
−3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
−3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
−3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
−3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
−3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
−3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Copyright © 2016. Pearson Australia Pty Ltd. All rights reserved.

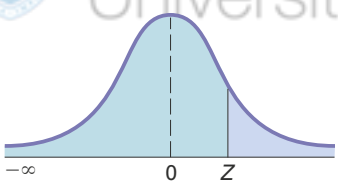
Stats for Mgrs Using Excel GE, Pearson Australia Pty Ltd, 2016. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/monash/detail.action?docID=5244245.
Created from monash on 2020-10-19 20:40:03.

MONASH University

TABLE E.2

The Cumulative Standardized Normal Distribution (*continued*)

Entry represents area under the cumulative standardized normal distribution from $-\infty$ to Z



Cumulative Probabilities										
Z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99897	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4.0	0.999968329									
4.5	0.999996602									
5.0	0.999999713									
5.5	0.999999981									
6.0	0.999999999									

Copyright © 2016. Pearson Australia Pty Ltd. All rights reserved.

Stats for Mgrs Using Excel GE, Pearson Australia Pty Ltd, 2016. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/monash/detail.action?docID=5244245.
Created from monash on 2020-10-19 20:40:03.